Short-answer bank and final pass

Practice questions · Day 24 · Mon 16 Nov · pack MXF

1. The 20 calculations that come up every year

Q1 territory (Ch1–2): power, AC circuits, three-phase

C1. Motor power, torque, efficiency and power factor

SPOT IT: you are given a rating (kW), the speed in rpm, and an efficiency or input V, I and P. You are asked for torque, losses, efficiency or pf.

METHOD:

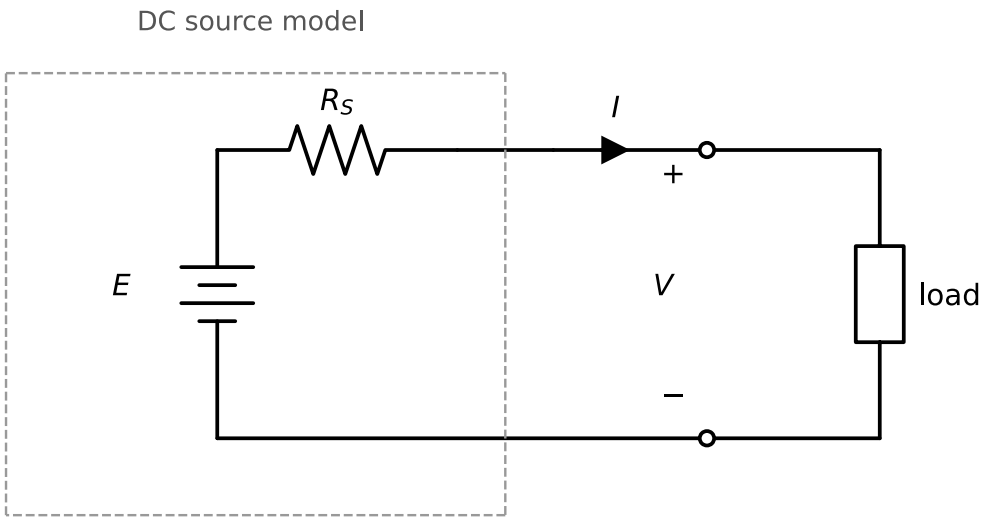
1. Convert the speed: $\omega = 2\pi n/60$.
2. Output power $P_{OUT} = \tau\omega$. The rating of a motor is its *output* power.
3. $P_{IN} = P_{OUT}/\eta$ and loss $= P_{IN} - P_{OUT}$.
4. Three-phase pf: $\cos\phi = P_{IN}/(\sqrt{3}V_L I_L)$.

KEY FORMULA: $\tau = \dfrac{P}{2\pi n/60}, \quad \eta = \dfrac{P_{OUT}}{P_{IN}}, \quad P = \sqrt{3}V_L I_L \cos\phi$

TRAP: using rpm instead of rad/s, or dividing the rating by efficiency to get the *output* power.

C2. DC source with internal resistance

SPOT IT: a battery or DC supply is given with an open-circuit voltage and a loaded voltage and current.



DC source model: EMF E behind RS, terminal voltage V = E - I RS

METHOD:

1. The open-circuit voltage is E, because no current flows and so there is no drop across R_S.
2. $R_S = (E - V)/I$ from the loaded reading.
3. Internal loss $= I^2 R_S$. Power to the load $= VI$.
4. New load: $V = E - IR_S$.

KEY FORMULA: $V = E - IR_S$

TRAP: forgetting that E is the *open-circuit* reading. Also, giving the power to the load as EI when it is VI.

C3. Star or delta resistive or impedance load

SPOT IT: "three identical X Ω loads on a V-volt 3ph supply, star then delta".

METHOD:

1. Star: $V_{ph} = V_L/\sqrt{3}$ and $I_L = I_{ph}$.
2. Delta: $V_{ph} = V_L$ and $I_L = \sqrt{3} I_{ph}$.
3. $I_{ph} = V_{ph}/|Z|$.
4. $P = 3I_{ph}^2 R$ (or $3V_{ph} I_{ph} \cos\phi$).

KEY FORMULA: delta power $= 3\times$ star power for the same impedances.

TRAP: putting V_L across a star phase. The quoted supply voltage is always the line voltage.

C4. Phasor circuit and complex power

SPOT IT: a series or parallel RLC circuit, with one phasor known (current or voltage). You are asked for the other phasors and for S, P and Q.

METHOD:

1. Write every element as an impedance: R, $+jX_L$, $-jX_C$.
2. Series: use the common current, so $V_k = IZ_k$ and $Z = \sum Z_k$. Parallel: use the common voltage, so $I_k = V/Z_k$ and $I = \sum I_k$.
3. Add the phasors in rectangular form and convert to polar at the end.
4. $S = VI^*$. Then $P = \operatorname{Re} S$ and $Q = \operatorname{Im} S$. A positive Q means the load absorbs reactive power, which is inductive.
5. For magnitudes only: in series RL, $|V| = \sqrt{V_R^2 + V_L^2}$, because the two phasors are at 90°.

KEY FORMULA: $S = VI^* = P + jQ, \quad Q = VI\sin(-\phi)$ where ϕ is the angle of I relative to V.

TRAP: adding magnitudes $(50 + 20)$ instead of phasors. Also, forgetting the conjugate, which flips the sign of Q.

C5. R and X from measured P, Q (or P, I, V)

SPOT IT: "a load draws I amps, absorbs P watts and Q VAr. Find series R and X, and L or C".

METHOD:

1. Series model: $R = P/I^2$ and $X = Q/I^2$.
2. If you only have V, I and P, use $|Z| = V/I$ and then $X = \sqrt{|Z|^2 - R^2}$.
3. Q > 0 means inductive, so $L = X/\omega$. Q < 0 means capacitive, so $C = 1/(\omega|X|)$.

KEY FORMULA: $R = P/I^2, \quad X = Q/I^2$

TRAP: using V^2/P, which is the *parallel* resistance, when the question asks for the *series* resistance.

C6. Power-factor correction with a capacitor

SPOT IT: a target supply pf is given. You are asked for Q_C or the capacitance.

METHOD:

1. Find the load P and Q, and draw the power triangle.
2. The capacitor does not change P, so $P_S = P_{load}$.
3. Find the target $Q_S = P_S \tan(\cos^{-1} \text{pf}_{target})$.
4. $Q_C = Q_S - Q_{load}$. This is negative, because the capacitor generates Q.
5. $C = |Q_C|/(\omega V^2)$. In three-phase, use the per-phase Q and the voltage across each capacitor.

KEY FORMULA: $|Q_C| = \omega C V^2$

TRAP: using the line voltage for star-connected capacitors. Also, quoting Q_C as positive when the question uses "absorbs" (the capacitor absorbs a negative Q).

Q2 territory (Ch3): magnetics and transformers

C7. Reluctance and inductance with an airgap

SPOT IT: a core with dimensions, μ_r, turns and a gap. You are asked for the reluctance(s) and L.

METHOD:

1. Convert everything to metres and m². The area of a circular section is $\pi d^2/4$.
2. $\mathcal{R}_g = l_g/(\mu_0 A)$ and $\mathcal{R}_c = l_c/(\mu_r \mu_0 A)$. If the core has infinite μ, then $\mathcal{R}_c = 0$.
3. Add them in series. One saw cut gives one gap. Two C-cores give two gaps.
4. $L = N^2/\mathcal{R}_{total}$.

KEY FORMULA: $\mathcal{R} = \dfrac{l}{\mu_r \mu_0 A}, \quad L = \dfrac{N^2}{\mathcal{R}}$

TRAP: cm² to m² is $\times 10^{-4}$, and mm to m is $\times 10^{-3}$. Also, the gap dominates even though it is tiny.

C8. Gap energy, force and DC current for a given B

SPOT IT: words like “energy stored in the airgap”, “force between the poles”, “hold the mass” or “DC current to give B”.

METHOD:

1. For DC current with infinite μ: $NI = Bl_g/\mu_0$. With finite μ: $NI = B(l_c/\mu_r\mu_0 + l_g/\mu_0)$.
2. Energy in the gap: $E = \frac{B^2}{2\mu_0} A l_g$.
3. Force: $F = \frac{B^2}{2\mu_0} A_{force}$. Two pole faces mean $A_{force} = 2A$.
4. To hold a mass, set $F = mg$ and solve for B.

KEY FORMULA: $E = \dfrac{B^2}{2\mu_0}\,\mathrm{Vol}, \quad F = \dfrac{B^2}{2\mu_0} A$

TRAP: missing the second gap or the second pole face, for example the paper in the 2024 test gives 2×0.07 mm of gap and 10 cm² of force area.

C9. Back-emf equation: turns, maximum voltage, B

SPOT IT: an AC winding with V and f, and either B_{max} or the turns given. You are asked for N, V or B.

METHOD:

1. $E = 4.44 N f \hat{B} A$, where E is rms and B is peak.
2. For N: $N = V/(4.44 f \hat{B}_{max} A)$, then round **up** to an integer, so that B stays below the limit.
3. Secondary turns: $N_2 = N_1 V_2 / V_1$, rounded to the nearest integer.
4. Wire resistance: $R = l/(\sigma \cdot \pi d^2/4)$.

KEY FORMULA: $V_{rms} = 4.44\, N f \hat{B} A$

TRAP: putting reluctance or μ_r in. They do not appear here: the voltage fixes the flux.

C10. Iron-loss scaling

SPOT IT: a total iron loss with a hysteresis and eddy split, then a new B and f.

METHOD:

1. Split the loss: P_H and P_E.
2. Use the course model: $P_H \propto B^2 f$ and $P_E \propto B^2 f^2$.
3. Scale each part separately, then add.
4. If only V and f change, then $B \propto V/f$.

KEY FORMULA: $P_{H2} = P_{H1}\left(\frac{B_2}{B_1}\right)^2 \frac{f_2}{f_1}, \quad P_{E2} = P_{E1}\left(\frac{B_2}{B_1}\right)^2 \left(\frac{f_2}{f_1}\right)^2$

TRAP: scaling the *total* with one law. The course formula sheet uses B^2 for hysteresis, not $B^{1.6}$.

C11. Ideal transformer and impedance referral

SPOT IT: an "ideal transformer, turns ratio a:1" with impedances on both sides.

Ideal transformer a:1 with a source impedance on the primary and a load Z on the secondary

METHOD:

1. $a = N_1/N_2$. Referring to the primary: $Z' = a^2 Z$.
2. Solve the single loop on the primary side.
3. Actual secondary values: $I_2 = aI_1$ and $V_2 = V_1/a$.

KEY FORMULA: $Z' = a^2 Z, \quad V_1/V_2 = a, \quad I_2/I_1 = a$

TRAP: using a instead of a^2, or referring in the wrong direction (to the low-voltage side, impedances get *smaller*).

C12. Transformer open-circuit and short-circuit tests

SPOT IT: "short-circuit test: V, I, P; open-circuit test: V, I, P". You are asked for the equivalent-circuit parameters.

Transformer tests: OC sees only RC parallel jXM, SC sees only the series R + jX

METHOD:

1. Three-phase: convert to per-phase first, with $V_1 = V_L/\sqrt{3}$ and $P_1 = P/3$.
2. Short-circuit test (series branch): $R = P_1/I_1^2$, $|Z| = V_1/I_1$, $X = \sqrt{|Z|^2 - R^2}$.
3. Open-circuit test (shunt branch): $R_C = V_1^2/P_1$, $Q_1 = \sqrt{(V_1 I_1)^2 - P_1^2}$, $X_M = V_1^2/Q_1$.

KEY FORMULA: short-circuit gives the series R and X. Open-circuit gives the parallel R_C and X_M.

TRAP: using series formulas on the open-circuit test, which gives a series R and X instead of the parallel R_C and X_M.

C13. Transformer on load: phasors, losses and efficiency

SPOT IT: a load in kW or MVA with a pf (or "generating Q"), at rated output voltage, using the simplified circuit referred to the primary.

Transformer simplified circuit referred to the primary: shunt branch at the input, series R1 + R2′ and j(X1 + X2′)

METHOD:

1. $V_2' = V_{L,rated}/\sqrt{3}\angle 0^\circ$ on the primary side.
2. $S = P/\text{pf}$ or $\sqrt{P^2 + Q^2}$. $|I_2'| = S/(\sqrt{3}V_L)$. The angle is $-\cos^{-1}\text{pf}$ for lagging and $+$ for leading (or when the load generates Q).
3. $V_1 = V_2' + I_2'(R + jX)$. The line voltage is $\sqrt{3}|V_1|$.
4. Losses: $P_{CU} = 3I_2'^2 R$ and $P_{FE} = 3V_1^2/R_C$. Then $\eta = P_{OUT}/(P_{OUT} + \text{losses})$, or $(P_{IN} - \text{losses})/P_{IN}$.
5. Open circuit: $I_1 = V_1/|R_C \parallel jX_M|$ and $P = 3V_1^2/R_C$.

KEY FORMULA: $V_1 = V_2' + I_2'(R + jX)$

TRAP: forgetting the factor of 3 on the losses. Also, forgetting $\sqrt{3}$ at the end when the question asks for the *line* voltage.

Q3 territory (Ch4–5 basics in part a, Ch7 induction machines)

C14. Poles, synchronous speed, ideal efficiency and a linear torque–speed curve

SPOT IT: "a 50 Hz AC machine with a rated speed of n rpm", or a fraction of rated torque.

METHOD:

1. $60f/n_R = p$ (pole-pairs). If it is not an integer, the machine is induction: round **down**, then poles $= 2p$ and $n_S = 60f/p$.
2. Ideal induction-machine efficiency $= n/n_S = 1 - s$.
3. Linear curve: the slip speed is proportional to torque. So $n = n_S - x(n_S - n_R)$, where $x = \tau/\tau_R$.
4. $P/P_R = x \cdot n/n_R$.

KEY FORMULA: $n_S = \dfrac{60f}{p} = \dfrac{120f}{\text{poles}}$

TRAP: p in the course formula is the number of pole-*pairs*. A 4-pole machine has $p = 2$.

C15. Induction-machine tests to parameters

SPOT IT: a DC resistance measurement, a no-load test and a blocked-rotor test on a star machine.

Induction machine tests: no-load sees RC parallel jXM, blocked rotor sees R1 + jX + R2′

METHOD:

1. Per phase: $V_1 = V_L/\sqrt{3}$ and $P_1 = P/3$.
2. Blocked rotor ($s = 1$, ignore the shunt branch): $R = P_1/I^2 = R_1 + R_2'$, so $R_2' = R - R_1$.

3. $X = \sqrt{(V_1/I)^2 - R^2} = X_1 + X_2'$. Split it equally if asked.
4. No load ($s \approx 0$): $R_C = V_1^2/P_1$ and $X_M = V_1^2/Q_1$.

KEY FORMULA: $R_2' = \dfrac{P_{BR}/3}{I_{BR}^2} - R_1$

TRAP: using the line voltage in $|Z|$. Also, forgetting to subtract R_1.

C16. Induction machine at a given speed (motor or generator)

SPOT IT: parameters, a speed, and "find I_1, P_IN, P_AG, τ". This is the Q3c staple worth 9–10 marks.

Induction machine simplified circuit: RC parallel jXM at the terminals, R1 + jX + R2′/s in series

METHOD:

1. Find n_S and $s = (n_S - n)/n_S$. If $n > n_S$, then $s < 0$ and the machine is a generator.
2. Compute R_2'/s, which is negative for a generator. Draw the circuit: $R_1 + jX + R_2'/s$, with the shunt branch across the terminals if it is not neglected.
3. Find $I_2' = V_1/(R_1 + R_2'/s + jX)$. Add the shunt current if it is included.
4. $P_{IN} = 3\,\mathrm{Re}(V_1 I_1^*)$ and $P_{AG} = 3I_2'^2 R_2'/s$.
5. $\tau = P_{AG}/\omega_S$. Check it with the torque formula. $P_M = (1-s)P_{AG}$.

KEY FORMULA:

$$\tau = \frac{3V_1^2\,(R_2'/s)}{\omega_S\left[(R_1 + R_2'/s)^2 + (X_1 + X_2')^2\right]}$$

TRAP: dividing by the *rotor* speed instead of the synchronous speed. Also, using the line voltage in place of V_1.

C17. Maximum torque and external rotor resistance

SPOT IT: "slip for maximum torque", "external resistance so maximum torque occurs at standstill", or sketching curves for added resistance.

METHOD:

1. $s_{\tau,max} = R_2'/|R_1 + jX|$.

2. Maximum torque at standstill means $s_{\tau,max} = 1$. So $R'_2 + R'_{EXT} = |R_1 + jX|$.
3. τ_{max} does not depend on R'_2. Adding resistance moves the peak towards standstill but keeps its height.

KEY FORMULA: $s_{\tau\max} = \dfrac{R'_2}{\sqrt{R_1^2 + X^2}}$

TRAP: using only X, or only X'_2, in the denominator. Also, drawing the higher-resistance curves with a higher peak.

Q4 territory (Ch8 and per-unit)

C18. Synchronous reactance from tests, and V, I in per-unit

SPOT IT: a table of excitation, open-circuit line voltage and short-circuit current, plus R. Then “delivering P (and Q or pf) at rated voltage”.

Synchronous machine OC and SC tests at the same field current

METHOD:

1. Use the same field current for both readings: $|Z_S| = (V_{OC,L}/\sqrt{3})/I_{SC}$.
2. $X_S = \sqrt{|Z_S|^2 - R^2}$.
3. $Z_B = V_B^2/S_B$ (line voltage, 3ph VA), so $X_S(\text{pu}) = X_S/Z_B$.
4. $V = |V|\angle 0^\circ$ pu (for example 0.97 for “3% below”). $S = P/\text{pf}$ or $\sqrt{P^2 + Q^2}$, divided by S_B.
5. $I = (S/V)\angle -\cos^{-1}\text{pf}$ when lagging or delivering Q.

KEY FORMULA: $Z_B = \dfrac{V_B^2}{S_B}, \quad I_B = \dfrac{S_B}{\sqrt{3}V_B}$

TRAP: forgetting $\sqrt{3}$ on the open-circuit *line* voltage. Also, dividing MW by MVA without first converting to S.

C19. Excitation, rotor angle and maximum power

SPOT IT: X_S in pu, $V = 1$ pu, a load at some pf. You are asked for E_0, δ and P_{max}.

Synchronous machine model: generator convention I out (top), motor convention Im in (bottom)

METHOD:

1. $I = (S/V)\angle \pm \cos^{-1}\text{pf}$, with $+$ for leading.
2. Generator: $E_0 = V + jX_S I$ (add R if it is given). Motor: $E_0 = V - jX_S I_{in}$.
3. $|E_0|$ is the excitation in pu, and $\delta = \angle E_0$.
4. $P = \frac{VE_0}{X_S}\sin\delta$, so $P_{max} = VE_0/X_S$ at $\delta = 90°$.

KEY FORMULA: $E_0 = V + jX_S I, \quad P = \dfrac{VE_0}{X_S}\sin\delta$

TRAP: using the wrong sign for the angle of a leading current. Leading output means the machine is under-excited, so E_0 is small and δ is large.

C20. Plant power factor with a synchronous motor

SPOT IT: induction motors (load kW, η, pf) plus a synchronous motor rated in kVA. You are asked for the best plant pf.

METHOD:

1. For each induction motor: $P = P_{mech}/\eta$ and $Q = P\tan(\cos^{-1}\text{pf})$, which is positive.
2. Synchronous motor: $P = P_{mech}/\eta$, and the most leading Q is $-\sqrt{S_{rated}^2 - P^2}$.
3. Add up all the P and all the Q.
4. $\text{pf} = \cos(\tan^{-1}(Q_{tot}/P_{tot}))$, and say lag or lead.

KEY FORMULA: $|Q|_{max} = \sqrt{S_{rated}^2 - P_E^2}$

TRAP: using the mechanical power in place of the electrical input power ($P_E = P_M/\eta$) in the capability formula.

2. Short-answer bank (Q4c style)

Past-paper short answers (official points)

S1. [Jun23 Q4c(i), 3 marks] Explain the concept of power-factor correction. Why is it useful in power systems?

S2. [Jun23 Q4c(ii), 3 marks] Carefully explain the differences and similarities between transformers and induction machines.

S3. [Jun23 Q4c(iii), 2 marks] A squirrel-cage rotor is made from a perfect conductor (infinite conductivity). How would this affect the machine?

S4. [Jun24 Q4c(i), 3 marks] Explain real, reactive and apparent power and their practical significance.

S5. [Jun24 Q4c(ii), 3 marks] Explain the main losses in a transformer, how they are modelled in the simplified equivalent circuit, and how they vary with output power.

S6. [Jun24 Q4c(iii), 3 marks] Why is the input current of an induction machine on no load not zero? What physical effects cause it?

S7. [2023 quiz Q7, 4 marks] How do you choose the number of primary turns for a given voltage and core? What happens with too few turns? With too many?

S8. [2024 test Q8, 4 marks] An ideal transformer drives a resistive load. What happens to the output voltage and current if the load resistance is doubled? How is a real transformer different?

S9. [Jun23 Q2c(iii), 2 marks] A transformer has a purely resistive load. Does the referred secondary voltage lead, lag or stay in phase with the primary voltage? Why?

S10. [Jun23 Q3a(ii), 1 mark] A 50 Hz AC machine has a rated speed of 560 rpm. Is it synchronous or induction? Why?

S11. [Jun24 Q3a(ii), 3 marks] How would you halve the no-load (synchronous) speed of a synchronous machine, a DC machine and an induction machine while keeping operation safe?

S12. [Jun24 Q2c(iii), 2 marks] A transformer is on open circuit and its input voltage is halved. By what factor do the input current, real power, reactive power and power factor change?

Likely new short answers

S13. [Ch7–8, 3 marks] Compare induction and synchronous machines.

S14. [Lecture 7.9.4–7.9.5, PQ short answer (f), 3 marks] Why is the voltage varied in proportion to frequency when an induction motor's speed is controlled?

S15. [PQ short answer (a), Lecture 7.4, 3 marks] How does the induction machine equivalent circuit differ from that of a short-circuited transformer, and why?

S16. [Ch3, Ch7, 3 marks] What is the effect of the airgap in a magnetic circuit, and in an induction machine?

S17. [Lecture 8.3, 3 marks] What are the advantages of the per-unit system?

S18. [Lecture 8.6.6–8.6.7, PQ short answer (k), 3 marks] Why is the power factor of industrial plants usually lagging, and why are synchronous motors sometimes used there?

S19. [Lecture 7.2.2, Jun24 Q3c, 3 marks] How can an induction machine operate as a generator? What are its limitations?

S20. [Lecture 7.1, 3 marks] Explain how an induction motor produces torque, and why it produces none at synchronous speed.

S21. [PQ short answer (b), Lecture 7.7, 3 marks] Explain the tests used to find the induction machine equivalent-circuit parameters.

S22. [PQ short answer (d), Lecture 7.9.1–7.9.3, 3 marks] Name the two types of induction-motor rotor and their main difference. What is the effect of extra rotor resistance?

S23. [PQ short answer (e), Lecture 7.8, 3 marks] Describe (in words) the torque–speed characteristic of an induction motor, naming the starting, rated and breakdown torques.

S24. [PQ short answer (g), Lecture 8.1–8.2, 3 marks] Using rotating magnetic fields, explain how a synchronous machine operates.

S25. [PQ short answer (h), Lecture 8.1.4–8.1.5, 2 marks] Why are salient-pole rotors used for low-speed generators and round rotors for high-speed generators?

S26. [PQ short answer (j), Lecture 8.6.5, 3 marks] What limits the reactive power capability of a synchronous machine?

S27. [Lecture 8.6.3–8.6.4, 3 marks] Explain the effect of the field current on a synchronous generator connected to a large power system.

S28. [PQ short answer (i), Lecture 8.1–8.4, 3 marks] Compare the synchronous machine equivalent circuit with that of a DC machine.

S29. [Ch3 iron loss, 2 marks] Why are transformer and machine cores laminated, and made from silicon steel?

S30. [PQ 9.1(d), Lecture 9.1, 2 marks] Why is electric power transmitted over long distances at high voltage?

S31. [Lecture 1.6.5, 2 marks] Why are large industrial machines three-phase rather than single-phase?

S32. [Lecture 7.1.1, 8.1, 2 marks] Why is a synchronous motor not self-starting on a fixed-frequency supply?